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Individualized Coaching to Improve Teacher Practice Across Grades and Subjects: New Experimental Evidence

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Abstract

This paper analyzes a coaching model focused on classroom management skills and instructional practices across grade levels and subject areas. We describe the design and implementation of MATCH Teacher Coaching among an initial cohort of fifty-nine teachers working in New Orleans charter schools. We evaluate the effect of the program on teachers' instructional practices using a block randomized trial and find that coached teachers scored 0.59 standard deviations higher on an index of effective teaching practices comprised of observation scores, principal evaluations, and student surveys. We discuss implementation challenges and make recommendations for researcher-practitioner partnerships to address key remaining questions.

Keywords: teacher coaching, professional development, randomized control trial, classroom management, instructional support

Individualized Coaching to Improve Teacher Practice Across Grades and Subjects: New Experimental Evidence

For over a century, school systems in the U.S. have attempted to improve instructional quality by investing in "on-the-job" teacher training. Today, 99% of public school teachers report participating in some form of professional development (Goldring, Gray, Bitterman & Broughman, 2013), with states and districts spending between $2,000 and $8,000 annually per teacher (Killeen, Monk, & Plecki, 2002; Miles, Odden, Fermanich, Archibald, & Gallagher, 2004; Picus & Odden, 2011). At the same time, research on professional development (PD) indicates that program quality is highly variable (Yoon, Duncan, Lee, Scarloss, & Shapley, 2007), with teachers themselves reporting mixed experiences (Farkas, Johnson, & Duffet, 2003). Impact evaluations also show that many PD programs fail to produce systematic improvements in teacher knowledge, behaviors, or effectiveness when implemented at-scale (Jacob & Lefgren, 2004; Glazerman et al., 2008; Yoon, et al., 2007). These findings are particularly troubling given the need to provide effective PD for teachers as districts adopt new teacher evaluation systems and the Common Core State Standards.

A growing number of districts and scholars have identified teacher coaching (Fletcher & Mullen, 2012) as an alternative to the short-term and generalized workshops that have characterized most PD programs (Darling-Hammond et al., 2009; Hill, 2007). Coaching programs commonly share several "critical features" including job-embedded practice, intense and sustained durations, and active-learning (Desimone, 2009). Experimental and quasi-experimental analyses of several coaching programs for kindergarten and early-elementary literacy and reading teachers have found that coached teachers became more effective instructors and that their students' academic achievement increased on standardized tests (Biancarosa, Bryk,

& Dexter, 2010; Marsh et al., 2008; Matsumura, Garnier, & Resnick, 2010; Neuman & Cunningham, 2009; Sailors & Price, 2010). At the same time, other training programs that incorporate coaching have not resulted in major changes in teacher practice or student achievement (Garet et al., 2008; Garet et al., 2011, Van Keer & Verhaeghe, 2005)

To date, few coaching programs have been developed to support the majority of teachers who teach subjects and grades other than early elementary literacy. One exception is the My Teaching Partner web-based coaching program, which focuses on improving the social, emotional, and instructional climates within an array of classrooms. In their experimental evaluation, Allen, Pianta, Gregory, Mikami and Lun (2011) found that bi-monthly coaching conversations over the course of an academic year increased achievement among secondary students by one fifth of a standard deviation in the year following the coaching intervention.

We contribute to the literature on teacher coaching by describing the design and implementation of MATCH Teacher Coaching (MTC), a coaching model focused on improving behavior management and instructional techniques across grades and subjects. We also present evidence of the causal effect of MTC on teachers' practices from the first cohort of teachers who participated a multi-cohort experimental study. In May of 2011, MTC recruited 59 early- to mid-career teachers in the Recovery School District in New Orleans to participate in a randomized trial of the year-long program. We then randomized teachers within schools to receive MTC coaching, in addition to any professional development opportunities their school provided, or to a status-quo control condition. MTC coaches worked with teachers to help them manage classroom behavior more effectively, use instructional time more productively, and align instruction to overarching curricular goals. After helping teachers identify areas for growth

during a week-long summer workshop, coaches provided ongoing, individualized feedback during four week-long coaching sessions throughout the year.

We utilize a rich set of qualitative and quantitative data to examine the implementation and effectiveness of the MTC program. Coaching logs and weekly summary emails written by teachers to coaches and school leaders allow us to describe the coaching model in detail and how it varied across individual teachers and over the course of the academic year. We triangulate the effect of coaching on teacher practices at the end of the coaching year and in the follow-up year as captured by three primary measures: classroom observations, principal evaluations, and student surveys. We also extend these analyses to explore whether coaching was equally effective for teachers across grade levels and subjects.

Background

Empirical Evidence on Professional Development

Despite a broad theoretical literature highlighting a clear causal chain connecting PD, teacher effectiveness, and student achievement (e.g., Desimone, 2009; Kennedy, 1998; Scher & O'Reilly, 2009; Yoon et al., 2007), a review of the empirical evidence on PD programs reveals mixed results. While some experimental and quasi-experimental studies find positive effects of PD on teaching practices and student outcomes (Connor et al., 2011; Landry, Anthony, Swank, & Monseque-Bailey, 2009; Penuel, Gallagher, & Moorthy, 2011; Powell & Diamond, 2011), others find null or mixed results (Cabalo, Ma, & Jaciw, 2007; Garet et al., 2008; Garet et al., 2011; Glazerman et al., 2008; Harris & Sass, 2011; Jacob & Lefgren, 2004; Santagata, Kersting, Givven & Stigler, 2011). Experts also note a lack of rigorous evidence on implementation fidelity and effects on proximal outcomes and intermediate mechanisms, such as individual teacher behaviors (Desimone, 2009; Wayne, Yoon, Zhu, Cronen & Garet, 2008). Most

importantly, there exists little evidence of PD programs impacting teacher practices and student achievement when taken to scale and applied across diverse contexts.

In light of these findings, scholars have sought to identify specific conditions under which PD programs might produce measurable improvements in teacher practice and student achievement. These discussions have led to a growing consensus that compartmentalized training sessions and school-wide workshops that characterize much of the PD provided to teachers are less effective than PD that is intensive, focused on discrete skill sets, and applied in context (Darling-Hammond et al., 2009; Garet et al., 2001; Hill, 2007; Wayne et al., 2008). Specifically, quantitative evidence suggests that programs with longer durations are more likely to be effective than shorter ones (Ramey et al., 2011; Yoon et al., 2007). Scholars also argue that successful PD cannot be divorced from teachers' own classroom contexts (Little, 2001). Instead, PD must approach teacher learning as a dynamic, active process where teachers may engage directly with student work, obtain direct feedback on their instruction, or review materials from their own classrooms (Desimone, Porter, Garet, Yoon, Birman, 2002; Garet et al., 2001).

Teacher Coaching as a New Model

Many scholars and practitioners have responded to these findings by re-envisioning PD in the form of teacher coaching. Coaching programs take a variety of forms, but most are centered on an individualized feedback process in which instructional experts work with educators one-on-one or in small groups to implement and improve specific aspects of teacher instruction (Fletcher & Mullen, 2012). Coaching cycles typically consist of classroom observations followed by targeted feedback about teachers' practices and specific recommendations for improvement. These cycles can occur frequently over the course of a full academic year or longer.

Coaching has gained its widest appeal among early elementary literacy and reading teachers through programs such as Reading First, the Literacy Collaborative, and Content-Focused Coaching. These programs pair the "critical features" of coaching described above with a deep content focus on literacy. Several experimental and quasi-experimental evaluations of these coaching models document improvements in teachers' literacy instruction and student performance on reading assessments (Biancarosa et al., 2010; Marsh et al., 2008; Matsumura et al., 2010; Neuman & Cunningham, 2009; Powell et al., 2010; Sailors & Price, 2010). However, two studies on the effect of professional development for early literacy that included coaching as one key component failed to find positive impacts that could be specifically attributed to coaching (Garet et al., 2008; Van Keer & Verhaeghe, 2005).

The limited research on content-specific coaching in other subject areas is mixed. A recent study found that two years of on-site coaching on mathematical content knowledge, pedagogy, and curriculum by trained mathematics coaches increased student achievement on standardized mathematics exams (Campbell & Malkus, 2011). A second study of a PD program for middle school mathematics teachers that included 18 days of follow-up coaching over the course of two years found no impacts on teacher knowledge or student achievement (Garet et al., 2011).

We are aware of only one other coaching model that is designed to develop teachers' non-content specific instructional practices. My Teaching Partner (MTP) (Allen et al., 2011) uses coaches trained in the Classroom Assessment Scoring System to assess videotaped lessons of teachers and facilitate conversations about the social, emotional, and instructional climates within classrooms. Experimental evidence among secondary school educators indicates that teachers randomly assigned to attend an initial workshop-based training and receive MTP web-

based coaching twice a month raised student achievement by 0.22 standard deviations on state standardized tests in the post-intervention year.

Directions for Current Research

Promising results from Allen et al. (2011) highlight the importance of studying additional coaching models focused on a broad array of practices that are relevant to teachers across grades and subjects. In particular, literature on effective teaching practices that draw on observations of teachers and student surveys highlights the importance of classroom management and general pedagogical practices (Kane & Staiger, 2012; Kane, Taylor, Tyler, & Wooten, 2010). School-level implementation of a classroom and instructional management program also has been found to support student achievement (Freiberg, Huzinec, & Templeton, 2009). While elements of general pedagogical practices are commonly incorporated into PD programming (e.g., Lemov's [2010] *Teach like a Champion* and Canter's [2010] *Classroom Management for Academic Success*), the research literature has yet to explore the potential of teacher coaching to improve these skills. Thus, we examine the question: What is the effect of an individualized coaching model focused on classroom management and general pedagogical skills on teachers' instructional practices?

The MATCH Teacher Coaching Model

We build on prior research by describing and presenting initial evidence on the effect of a time-intensive, individualized coaching program focused on improving teachers' classroom management and general pedagogical practices. The Match Teacher Coaching model grew out of the teacher training and professional development programming developed at MATCH Charter Public Schools in Boston, MA. The model consists of three main components: a set of core materials and resources, a 4-day summer training institute, and one-on-one coaching during the

academic year. Below we describe the MTC model as it was first implemented in New Orleans Charter schools during the 2011-2012 academic year.

MTC is centered around a set of instructional practices developed by a range of expert practitioners and a mindset that focuses on growth and continual improvement. Core texts assigned to teachers as part of the coaching program include books on specific pedagogical practices and class management techniques by Doug Lemov (2010), Fred Jones (2007), and Lee Canter (2006) as well as a set of original MTC booklets on classroom management, developing relationships with students and parents, and executing effective lessons. In addition, teachers read work by Carol Dweck (2006) on the powerful effects of adopting a growth mindset toward professional development and students' learning potential.

Participating teachers attend a four-day training workshop during the summer led by MTC coaches. The training provides a total of 21 hours of content, with 8 hours set aside for reading and reviewing assigned materials. Teachers are divided up into two different groups for the entire training based on coaches' assessments of their instruction from observations conducted in the spring, principals' survey responses, and teachers' own self-assessments. One group focuses primarily on developing classroom management techniques and building relationships with students and parents, while the other group focuses on lesson planning and execution. Each day coaches introduce new material, view and discuss example videos of techniques with teachers, and give teacher opportunities to practice new skills. The goals of the training are to develop an individualized coaching plan and to help teachers acquire a common language around instruction and prepare to start the year with a clear set of classroom norms and routines. Coaches and teachers work together to develop a coaching plan which identifies 3-5 instructional goals for the year and a set of specific action-steps to achieve these goals. Common

goals for the academic year include improving behavior management, classroom climate, lesson planning and execution, productive use of class time, and student engagement.

Coaches then work with individual teachers one-on-one throughout the course of academic year to support teachers in achieving their goals. Coaches and teachers work together during four week-long coaching sessions, which allow for repeated rapid cycles of observation, feedback, implementation, and reassessment. Each day coaches observe teachers for at least half the day and then meet with them during planning periods or after school to debrief. Teachers are evaluated through formative assessments on the classroom observation rubric developed by the coaching program. Coaches also help teachers to identify daily and weekly goals, assess the extent to which teachers met these goals, and use these assessments to identify future growth areas.

Debrief sessions are structured into five stages. Coaches begin by reviewing the goal and main takeaway from the previous debriefing session. Takeaways consist of a specific aim, prescriptions for techniques a teacher can use to meet this aim, and plans for preparing and practicing to implement these techniques. Coaches then describe effective practices they observed the teacher using and areas where the teacher has room for improvement. The coach and teacher work to develop a new main takeaway and conclude with the questions and other small suggestions a coach has that are not central to the main goal. At the end of each week-long coaching session, teachers send an email to their coach and to their school principal outlining the goals addressed that week, plans for implementing feedback from their coach, and goals for future sessions. Between week-long coaching sessions, teachers communicate with coaches about their progress via email or phone every one-to-two weeks.

The MTC coaching program in New Orleans was delivered by three coaches in 2011-2012, all former teachers in urban public schools with professional experience in education non-profits, and charter school management organizations. Two coaches were female and one was male. Coaches' ages ranged from the late twenties to early forties. Two of the coaches held master's degrees in education. We provide data and further details about how MTC was implemented in our findings section below.

Research Design

Sample

MTC coaches worked in partnership with New Schools for New Orleans to recruit teachers employed at charter schools across the Recovery School District. The Recovery School District is a statewide district in Louisiana formed in 2003 to transform underperforming schools, the vast majority of which are in New Orleans. Teachers of all grade levels and subject areas were eligible to participate. Recruitment efforts focused on early- and mid-career teachers, a population known to require on-site support and assistance (Kaufman, Johnson, Kardos, Liu, & Peske, 2002). Based on principal nominations and word-of-mouth, 91 teachers expressed some level of initial interest in the program. Given capacity constraints, MTC staff chose to limit the pool of teachers who would be eligible to receive coaching to those teachers who expressed high levels of interest in the program, completed all required paperwork, and received permission from their principal. This restriction resulted in a final sample of 59 teachers: 33 elementary school teachers, 16 middle school teachers, and 10 high school teachers. Twenty-five teachers taught all core subjects in self-contained classrooms, 21 taught in the humanities (English Language Arts (ELA) or social studies), and 13 taught in STEM fields (science or mathematics).

In Table 1, we present descriptive statistics for participating teachers and those not selected for participation. Compared to the general population of public school teachers in the United States, participating teachers were much more likely to have entered the profession through alternative licensure programs (76% vs. 24%), such as Teach for America or TeachNOLA, and to be African American (34% vs. 6%) (Goldring, Gray, Bitterman, & Broughman, 2013). Over three-fourths of participating teachers attended an undergraduate institution whose admissions process is rated as "Very Competitive" or higher by Barron's rankings. Including the coaching year, 27% of teachers were in their first or second year, 42% were in their third or fourth year, and 31% were in their fifth year of teaching or higher. Compared to those teachers not selected into the program, study participants had a higher level of initial interest, by design, and were more likely to be white. However, participants were similar to non-selected teachers in their gender, experience, and certification pathway. As we discuss in our conclusion, the high level of interest among participating teachers may be critical to the potential success of the coaching program.

Participating teachers taught across 20 different charter schools operated by 16 different charter management organizations. These schools included seven elementary schools, eight K-8 schools, three middle schools, and three high schools. All schools in which coaches worked served student populations that were over 90% African-American; in all but one, over 90% of students were eligible for free- or reduced-price lunch. School rankings on a state "performance index" ranged from 62 to 113 with an average of 82, slightly higher than the Recovery School District average of 74, but notably lower than the state average of 99.

Experimental Design

Among the 59 participating teachers in this first cohort, half were randomly selected to be offered coaching using a block randomized design. Randomized control trials are the "gold standard" for estimating causal effects and made sense in our context where the demand for coaching exceeded MTC's capacity to supply coaching to all interested teachers. We randomized teachers within the schools they taught at during the 2010-11 school year. This decision was necessary in order to recruit schools to participate by guarantying every principal that half of the teachers they nominated would receive coaching. A simple randomized design might have resulted in some schools having all their teachers randomized to the control group.

Implementing this study as a blocked randomized trial had both important advantages and drawbacks. First, this design was a critical condition for recruitment. Second, it ensured that any treatment effect would not be confounded by the dominant effect of teachers at one or two schools should a majority of those teachers end up in the treatment or control condition due to sampling idiosyncrasies. Third, assigning treatment at the teacher level, rather than at the school level, greatly increases our statistical power. These advantages come at the cost of an inability to fully leverage peer support networks among all participating teachers within a school, as well as potential spillover effects between coached and control-group teachers in the same school (Wayne et al., 2008). While spillover has the potential to downwardly bias estimates, research suggests that spillover would have to reduce treatment effects by upwards of 60% before a cluster randomized design produced greater statistical power than a block randomized design (Rhoads, 2011).

Teachers in the control group did not receive any form of support from the MTC program. Although staff development practices varied across the charter schools who participated in this study, no schools provided formal coaching to teachers at the time. Informal

discussions with participating teachers suggest that professional development activities were very limited and generally ineffectual. Thus, the treatment-control contrast represents a substantial departure from the informal and infrequent professional development activities that were the status quo.

We examine a range of baseline measures to confirm the validity of our randomization process by comparing the demographic characteristics of teachers assigned to treatment and control groups. The results reported in Table 1 provide strong evidence that groups were balanced on observable characteristics after randomization. Differences in mean values of observable teacher characteristics across the treatment and control groups are small and insignificant for each measure.

Data and Measures

Two sources of qualitative data allow us to assess fidelity of implementation and examine the content and methods used during coaching sessions. First, we examined emails that teachers sent to their coaches and school leaders outlining which classroom practices they worked on in a given week. This activity was required of teachers as part of the coaching cycle. We also analyzed coaching logs that coaches completed after each debriefing session. As above, this process was required of coaches and monitored by the research team. In these logs, coaches identified which tools from a list set at the beginning of coaching they used when working with each teacher. This list included providing direct feedback to teachers, lesson planning, tweaking classroom management plan, collecting data, watching a video of instruction, and reviewing action steps. Coaches also could write in additional tools that were not included in this initial list.

Given our primary goal of investigating whether a generalized coaching program can be effective across the full range of grades and subjects, we focus our analyses on measures of

teachers' instructional practices common across K-12 classrooms. These primary sources of data include a classroom observation rubric developed by MATCH, a principal evaluation form based on previous studies, and the TRIPOD student survey. In order to mitigate the likelihood of Type I error due to multiple hypothesis testing (Schochet, 2008), we pre-selected a parsimonious set of five measures from these data as our confirmatory outcomes. Following Anderson (2008) and Kling, Liebman and Katz (2007), we also construct a summary index of these measures to guard further against Type I error.

MATCH classroom observation rubric. The MATCH observational rubric version 1.0 was developed by leaders at MATCH Public Charter School to provide both formative and evaluative feedback around instruction (see Appendix A for full instrument). The rubric is comprised of two overall codes, *Achievement of Lesson Aim* and *Behavioral Climate,* each with a set of key indicators. For *Achievement of Lesson Aim*, these include clarity and rigor of the aim, alignment of learning activities with the lesson aim, and assessment and feedback. For *Behavioral Climate,* indicators include time on task, transitions, and student responses to teacher corrections. While viewing instruction, raters take notes on each of these indicators and the extent to which they are present in a teacher's instruction; in particular, raters provide examples from the lesson. At the end of the lesson, raters score each of the two main codes holistically on a scale of 1-10 based on these pieces of evidence. Score ranges are aligned to standards set by MATCH for high-quality instruction: 1-3 = Below Expectations, 4-5 = Approaching Expectations, 6-7 = Meets Expectations, 8-10 = Exceeds Expectations.

Coaches observed and rated teachers on the rubric in the spring semester prior to randomization. In the following two spring semesters (i.e., at the end of the coaching year and in the follow-up year), experienced outside observers who were blind to treatment status observed

and rated a class taught by each teacher on two separate occasions (one rater at each occasion).[1] After receiving training on how to use the instrument, raters achieved between 80% and 100% one-off agreement rates with the director of MTC for both dimensions in each year. One-off agreement rates also are used by Bell et al. (2012) in assessing reliability of the Classroom Assessment Scoring System (CLASS), which has a similar scale to the MATCH rubric.

Another statistic often used for observation instruments is the intraclass correlation, which describes the amount of variation in teacher-level scores that is attributable to the teacher, as opposed to the rater(s) who observed the teacher or the specific lesson(s) observed. However, in this context, we are less concerned with rater, or lesson, variance given that any variance in scores not attributable to teachers will be balanced across treatment and control groups. This is ensured by the fact that each rater observed all teachers and that raters typically observed all teachers in a given school (both treatment and control) in the same week. We create teacher scores for each code by averaging raw scores across our two raters and then standardizing average scores to be mean zero and standard deviation one within each time period (i.e., baseline, spring, follow-up).

Principal survey. We utilize a principal survey in which we combine and slightly adapt survey items developed by Jacob and Lefgren (2008) and Harris and Sass (2009). Both studies provide evidence of the predictive validity of these items by documenting the correlation between a composite survey measure and teacher value-added scores in math and reading (0.32 and 0.29 respectively for the former survey, and 0.28 and 0.22 for the latter). We asked school administrators (e.g., principal, direct supervisor) who were most directly responsible for teachers' supervision to complete the survey. These administrators rated teachers on a scale from

[1]In an analysis of a range of other observational instruments, the Measure of Effective Teaching Project found that this scoring design, two observations of approximately 45 minutes with each done by a different observer, produced a reliability of 0.67 with school administrators as raters (Kane & Staiger, 2012).

one (inadequate) to nine (exceptional) across 10 items: *Overall Effectiveness, Dedication and Work Ethic, Organization, Classroom Management, Time Management in Class, Time on Task in Class, Relationships with Students, Communication with Parents, Collaboration with Colleagues,* and *Relationships with Administrators.* We also asked principals to rank teachers in a given quintile of effectiveness compared to all the teachers at their school. Principals completed this for each teacher in the spring prior to the coaching year and again the following two springs after the experiment was concluded. We create a composite score of teachers' overall effectiveness, *Principal Evaluation Composite*, by standardizing individual items within each year, averaging scores across all 11 items above, and then re-standardizing this composite score to be mean zero and standard deviation one within each time period. We estimate an internal consistency reliability of 0.91 or greater in all three administrations.

It is important to note that it was not feasible to keep principals blind to teachers' experimental condition. This could potentially bias principal evaluations scores if, for example, principals were inclined to rate teachers who participated in coaching more favorably. Although there appeared to be no incentive for principals to rate coached teachers more favorably given that their ratings were for research purposes only and were not used in any formal teacher or school evaluations, we cannot definitely rule out this potential threat.

TRIPOD student survey. The TRIPOD survey is comprised of items designed to capture students' opinions about their teacher's instructional practices. Measures of teacher effectiveness are categorized into seven domains or "C's": *Care, Clarify, Control, Challenge, Captivate, Confer,* and *Consolidate,* each with an internal consistency reliability of 0.80 and above (Kane & Staiger, 2011). Students in grade three and higher rated each item on a five-point Likert scale. A member of the research team administered the survey to early-elementary

students by reading survey items aloud in small groups and asking students to select among three responses: no, maybe and yes. Students completed the survey once in the spring of the coaching year, as well as in the follow-up year.

We focus our confirmatory analysis on two specific measures, *Control* and *Challenge*, which ask students about the behavioral climate and the level of academic rigor in their class. In addition to being best aligned to the coaching program, these two measures were found to be most predictive of teachers' ability to raise students' test scores in both math and reading, with correlations of 0.22 and 0.14, respectively (Kane & Staiger, 2011). Following the practices of the TRIPOD project, we derive scores for each of the 7C's by rescaling items so that they are consistent across all forms, standardizing Likert-scale response options for each item, and calculating the mean response across items. We then standardize teachers' average score for each of the 7C's to be mean zero and standard deviation one within each time period. For illustrative purposes, we also examine the proportion of students who agreed with a single item from the *Consolidate* domain, "In this class, we learn a lot every day".

Summary index. We create an index of *Effective Teacher Practices* by taking a weighted average of the five measures described above – the two MATCH rubric items, the principal survey composite, and the two TRIPOD composites – such that all three data sources are given equal weight (i.e. 1/3 outside observer ratings, 1/3 principal evaluation scores, and 1/3 student survey ratings). We then standardize the index to be mean zero and standard deviation one.

Data Analyses

Implementation analyses. In order to assess fidelity of implementation, we first coded emails that teachers sent to their coaches and principals at the end of each week of coaching. To do so, we generated a list of all terms that teachers used to describe activities in which they

engaged as well as key focal areas. Then, we sorted these into broader categories that aligned with the stated goals of the MTC program (e.g., behavior management, instructional delivery). We found that teachers and coaches generally used a common language and set of terms in these emails developed during the summer training sessions and reinforced by the training materials. In order to ensure reliability, the same rater conducted all coding.

Next, we compiled data from coaching logs to describe the tools used during debriefing sessions. Given that coaches generally chose from a list of tools, this process was relatively straightforward. When coaches wrote in additional tools, we created a separate category for these. Coaching logs also captured detailed records of the specific coach that each teacher worked with during a given coaching session and the total number of weeks of coaching each teacher received throughout the academic year.

Treatment effect analysis. We estimate the effect of MTC on our outcomes of interest using Ordinary Least Squares (OLS) and multilevel regression. We analyze our teacher-level measures including observation scores, principal ratings, and teacher self-evaluations by fitting the following OLS model where *Y* represents a given outcome of interest measured at the end of the coaching year for teacher *j* in school *s* at time *t*:

$$Y_{jst} = Y_{j,t-1} + \beta\, MTC_j + \alpha_{s,t-1} + \varepsilon_{jst} \qquad (1)$$

For each of these teacher-level outcomes, we are able to include a baseline measure, $Y_{j,t-1}$, to increase the precision of our estimates. We include fixed effects for the schools where teachers taught at the time of randomization, $\alpha_{s,t-1}$, to account for our block randomized design. We omit random effects for the schools where teachers worked during the coaching and follow-up years in all models because they are highly collinear with our blocking indicators since only three teachers switched schools between the time of randomization and the beginning of

coaching; nine teachers who participated in the follow-up analysis switched schools.[2] However, we cluster our standard errors at the school-level in the current year. For analyses that examine outcomes measured in the follow-up year, we retain the same control for baseline measures and blocking indicators captured prior to randomization. The subscripts on these covariates are thus *t*–2. We extend these analyses by examining heterogeneity in program effects on instructional practices across grade-levels, subject areas, and schools to shed light on whether the program was equally effective across settings.

We analyze our student-level survey responses by fitting an analogous multilevel model where students, *i*, are nested within classrooms, *c*, and teachers, *j*:

$$A_{ijcs} = \beta\, MTC_j + \alpha_{s,t-1} + (\nu_j + \varphi_c + \epsilon_{ijcs}) \qquad (2)$$

We include blocking indicators, as above, and random effects for classrooms, φ_c, and teachers, ν_j, and cluster our standard errors at the school level in the current year.

In both models, the coefficient β on the indictor for whether a teacher was randomly offered the opportunity to participate in MTC is our parameter of interest. We interpret these estimates as Average Treatment Effects given that every teacher offered coaching, except two who withdrew prior to the 2011/12 school year, fully participated in the program. These two teachers who were offered coaching do not have the necessary data to be included in our analysis given that one left teaching and another switched schools and withdrew from the study.

Findings

Coaching Implementation, Content, and Techniques

Overall, the MTC program was implemented with close fidelity to the original coaching plan. Every teacher who participated in the coaching program received four or five total weeks of

[2] Likelihood ratio tests comparing models with and without school random effects fail to reject the null hypothesis that these models are statistically significantly different (p=0.99).

coaching, with 24 teachers receiving the planned four weeks and four teachers receiving an extra fifth week of coaching. Coaches reported that variation in the number of weeks of coaching that teachers' received was primarily due to coaches decisions about which teachers needed additional support. Twenty-one teachers worked with the same coach throughout the academic year, while seven worked with two coaches at least once due to scheduling difficulties when making-up cancelled coaching sessions. Coaches estimated that the average contact with an individual teacher was roughly 50 hours over the course of the school year, meeting program goals.

We analyzed teachers' emails and coaches' logs to assess the content and techniques used during coaching sessions. Five broad focus areas emerged from these data: behavior management, classroom climate, instructional practices, productivity, and student engagement (see Appendix Table A1 for examples of activities from each). For example, some teachers who focused on *behavior management* worked on implementing a consequence/reward system or monitoring students by moving throughout the classroom; some teachers who focused on *instructional practices* worked on aligning activities to the overall lesson aim and on writing exit tickets to assess student understanding of the lesson aim. In Figure 1 Panel A, we show the proportion of total week-long coaching sessions in which each focus area was addressed, as well as the proportion of teachers who ever worked on a particular focus area. Because teachers worked on multiple areas in a given week, proportions do not sum to one. Over the course of the academic year, teachers focused predominantly on behavior management and instruction, with 62% of all sessions covering the former and 59% the latter. Ninety-three percent of teachers received coaching on behavior management and instruction during at least one week-long coaching session.

The degree to which teachers were coached in these two areas varied widely across teachers, depending on their specific needs. As illustrated in Figure 1 Panel B, some teachers never worked on behavior management or concentrated on it for only one week, while others spent most, or even all, of their coaching sessions on management issues. Over the course of the year, many teachers who began with classroom management issues shifted focus toward instruction. In week one, 37% of sessions focused on management and 23% on instruction. By week four, these percentages had largely reversed to 20% and 32%, respectively.

We also found that coaches used a variety of coaching techniques but relied heavily on a few central practices. The most common practice was providing teachers with direct feedback about what they could do better or differently in future lessons, something that occurred in 78% of all coaching sessions. The second and third most common techniques were lesson planning with teachers and reviewing digitally recorded lessons, used in 52% and 38% of all sessions. Daily coaching cycles often incorporated these practices in combination where coaches would review a recorded lesson with a teacher, give them direct feedback about the lesson, and then work with them to plan ways to incorporate this feedback into their upcoming lessons.

The Effect of Coaching on Teachers' Practices

Coaching year. Simple descriptive statistics of changes in teachers' effectiveness over time as judged by outside observers, principals, and the teachers themselves all illustrate greater gains for teachers who received coaching compared to those randomly assigned to the control group. On average, treatment-group teachers improved 1.26 and 1.47 scale points (on a 10-point scale) more on the MATCH rubric domains, *Achievement of the Lesson Aim* and *Behavioral Climate*, than control-group teachers at the end of the coaching year. Principals rated teachers who received coaching as improving 0.31 scale points (on a nine-point scale) more than their

control-group counterparts on the *Principal Evaluation Composite*. MTC teachers' assessments of their own effectiveness show average gains of 0.71 scale points (on a nine-point scale) on a composite measure made up of the same items from the principal survey composite, while control-group teachers rated themselves no differently from fall to spring.

Using our full regression framework which only compares treatment teachers to control-group teachers in the same randomization block, we found that the MTC program improved teachers' effectiveness across a range of practices. As shown in Table 2, MTC teachers scored 0.59 sd higher than control group teachers (p=.024) on our *Effective Teacher Practices* index consisting of observation scores, principal evaluations, and student surveys. As a relative benchmark, this improvement is almost 50% larger than the average difference on the same index between teachers in their first or second year and the more experienced teachers in our sample (0.44 sd, p=.156), controlling for treatment status. Across individual instruments, our estimates of the effect of MTC on teachers' practices are consistently positive but imprecisely estimated, with several estimates achieving only marginal significance at the 10% level. Trained classroom observers rated coached teachers 0.58 sd (p=.079) and 0.66 sd (p=.049) higher on *Achievement of Lesson Aim* and *Behavioral Climate*, respectively. Principals rated teachers who received coaching 0.29 sd (p=.099) higher on the *Principal Evaluation Composite*. We interpret this marginally significant result with additional caution given the potential bias in principals' ratings.

These changes in coached teachers' practices had mixed effects on the classroom experiences of their students. Similar to outside observers and principals, students rated teachers who received coaching as more effective at challenging them with rigorous work. Specifically, coached teachers scored 0.31 sd (p=.007) higher on the *Challenge* domain of the TRIPOD survey

than control group teachers. However, we found no statistically significant effects on the *Control* domain, a measure of students' perceptions of the teachers' classroom management skills. We illustrate the magnitude of these effects by estimating the impact of MTC as measured by a single item. Students of the teachers who received coaching were eight percentage points (p=.018) more likely to agree that "In this class, we learn a lot almost every day."

Follow-up year. We also examined the effect of MTC on teachers' effectiveness in the year following the end of coaching to assess whether teachers continued to benefit from coaching even though coaches no longer supported their instruction. It could be that coaching effects fade out with time or that they increase as teachers are able to leverage their new skill sets starting on the first day of class. In the follow-up year, we were able to re-recruit 33 of the 42 teachers in our sample who returned as classroom teachers. The high rate of teacher turnover in our sample, 28.8%, is reflective of the 27% annual turnover rate among teachers across the Recovery School District in the 2011/12 school year (Cowen Institute, 2012). All fourteen original randomization blocks were represented in this follow-up sample, indicating that no single school or subset of schools drove participation rates. Seven of these included at least one teacher from both the treatment and control groups.

We found that despite this high attrition rate, there were no statistically significant differences in observable characteristics between our full sample and those teachers who remained in the study for a second year (See Appendix Table A2). We also show that characteristics of the treatment group teachers who participated in the follow-up year were similar to those of the 12 teachers in control group, on average. We do observe a marginally significant difference among teachers' initial interest. Given the high rates of attrition, we interpret our post-coaching year estimates as suggestive rather than strong causal estimates.

In Table 2 column 4, we present estimates of the effect of coaching on teachers' classroom practices in the post-coaching year. Our estimate of the effect of coaching in the follow-up year on our *Effective Teacher Practices* index (0.476 sd p=.364) is quite similar in magnitude to the effect at the end of the coaching year, but is indistinguishable from zero in our smaller sample. This finding is far from conclusive, but at least suggests that coached teachers were able to sustain many of the improvements they had made even when they no longer received the support of MTC coaches.

Heterogeneity in Treatment Effects on Teachers' Practices

In addition to the average treatment effects presented above, we explored whether coaching was equally effective across grade-level and subject taught. These analyses help to shed light on the degree to which our estimates are generalizable across the grades and subjects represented on our sample. We focused these analyses on outcomes measured in the coaching year given our larger sample and the similarity in estimates between the coaching and follow-up years. In Table 3, we report results from models where we have replaced our single treatment indicator with sets of treatment indicators across subgroups of teachers. Our estimates of the effects of MTC on the index of *Effective Teacher Practices* are uniformly positive and of relatively similar magnitude across subgroups of teachers. We did find some suggestive evidence that coaching may have been more effective for teachers in STEM fields; however, we lack the statistical power to detect whether or not coefficients across subgroups are statistically significantly different from each other. Overall, it does not appear that there were substantial differences in the effect of MTC on teachers who teach different grade levels and subjects.

Our randomized block design also allowed us to explore variation in treatment effects across schools, which has important implications for the generalizability of program effects

across settings. We estimated school-level variance parameters by modifying models (1) and (2), exchanging fixed effects for prior-year school blocks for random effects, and including an interaction term between our treatment indicator and these prior-year school random effects (Raudenbush & Liu, 2000). In Table 3, we report the standard deviation of the variance in treatment effects as well as the *p*-value associated with a Likelihood Ratio test of the significance of our prior-year school-by-treatment random effects.

Two interesting patterns emerge from this analysis. First, coaching effects on measures of teacher practice that are more broad (i.e, our *Effective Teacher Practices* index and the *Principal Evaluation Composite*) are relatively consistent across schools. Conversely, we observe substantial variation in treatment effects for measures of teachers' practices that are more specific in nature (i.e., *Achievement of the Lesson Aim, Behavioral Climate*, *Control, Challenge*).[3] These findings make sense given the individualized nature of the coaching program, where teachers received coaching in different areas depending on their specific needs.[4]

Threats to Validity

Attrition and Missing Data

We examined the robustness of our confirmatory analyses to sample attrition and missing data in several ways and found that the character of our results unchanged. During the coaching year, seven teachers in our study did not have the necessary data to be included in our analysis. Two of these were treatment teachers; one dropped from the study because she left the district and the teaching profession prior to the beginning of the school year, and the other

[3] We conduct parallel analyses using a fixed effect framework and find that our results are consistent with these findings. In this approach, we maintain our prior-year school blocks as fixed effects and replace our generalized treatment indicator with school-specific treatment indicators.

[4] We interpret our estimates of the variation in treatment effects for *Achievement of the Lesson Aim* and *Behavioral Climate* as indicative of true heterogeneity given that our failure to reject the null hypothesis is likely due to our limited statistical power for this test. Unlike statistical power for average treatment effects in block randomized trials, power for detecting variation in treatment effects are driven largely by the number of observations per school rather than the number of schools (Raudenbush & Liu, 2000; Konstantopoulos, 2008).

dropped because he switched schools and chose to withdraw from the study. Of the five control teachers who were missing data, four left teaching and one decided not to participate in the study or data collection. Between the coaching and follow-up year, an additional 19 teachers attritted due to turnover out of New Orleans or out of teaching, or because they chose not to participate.

We first explored patterns of attrition by examining whether the relationship between the probability of attritting and observed demographic characteristics differ across teachers in the treatment and control groups. If less effective treatment-group teachers or more effective control-group teachers were censored from the study, our results would be biased upwards. To explore this potential source of bias, we regressed each demographic characteristic on an indicator for attritting, an indicator for coached teachers, and their interaction. In Table 4, we report the parameter estimates associated with these interaction terms, which test for differential attrition, for both the coaching and the follow-up years. For the coaching year, we found no evidence of differential attrition across any of the observed teacher characteristics, suggesting that those teachers who were censored were not systematically different across the treatment and control groups. For the follow-up year, we noted substantively different rates of attrition between the treatment and control groups. We also found some evidence that control-group teachers who attritted had lower levels of initial interest than coached teachers who did so. However, we found no difference in any other observable characteristic between coached and control-group teachers who were censored from the study in the follow-up year.

We used two primary approaches to test the robustness of our findings to attrition and missing data. First, we followed Kling, Liebman and Katz (2007) by imputing baseline and outcome means within each experimental group and re-estimating our results in the full sample. By imputing group means, we have assumed that missing data is missing completely at random.

We relaxed this strong assumption in our second approach by using multiple imputation, which assumes that data are missing at random, conditional on the observed characteristics and ratings of teachers that we do have in our data (Rubin, 1987). We implemented this approach by imputing missing data for baseline and outcome measures of effectiveness using teacher characteristics presented in Table 1 and an indicator for treatment status. Because both techniques assume some level of randomness in missing data, we do not present results from these strategies for the follow-up year. In this second year, sample attrition may not be independent from treatment status despite the lack of notable differences in observable characteristics across groups.

In Table 2, we report results from each of the methods described above alongside our original estimates. We found that estimates of MTC effects are largely consistent with our primary analyses when we used mean imputation (column 2) with the exception of the *Principal Evaluation Composite,* which was attenuated and no longer marginally significant. In column 3, we present the average point estimates across 10 imputed datasets as well as their associated standard errors derived from standard formulas. Again, our results were largely unchanged. Overall, we interpret these findings as strong evidence that our estimated effects in the coaching year cannot be explained away by differential sample attrition across experimental groups.

Spillover Effects

Given our design in which teachers were randomized to either the treatment or control group within schools, it is possible that control-group teachers were exposed to elements of coaching through their colleagues. Analyses of end-of-study teacher surveys indicated that nine of the 24 control-group teachers who remained in the study did learn about instructional techniques taught by coaches from their colleagues who received coaching. Seven of these

teachers reported using these new techniques in their own classrooms. In addition, coaches indicated that several principals incorporated coaching techniques into their school-wide PD. These data suggest that our estimated treatment effects likely understated the full effect of the MTC program. The adoption of instructional techniques taught by MTC provides further evidence of the efficacy of these practices.

Discussion and Conclusion

The Challenge and the Evidence

A growing consensus is emerging among policymakers and scholars that teachers, and teaching quality, should be a focal point of any large-scale effort to improve public education. Efforts to improve the quality of the teacher workforce through selective recruitment and retention are limited by the sheer scale of the education sector and our relative inability to predict who will be an effective teacher (Clotfelter, Ladd, & Vigdor, 2007; Rockoff, Jacob, Kane, & Staiger, 2011). The challenge, then, is how to improve the instruction of the 3.5 million teachers in classrooms across the United States. This is not a new challenge but rather a persistent one. Schools invest billions of dollars annually in programming, personnel, and support services intended to promote professional growth among teachers (Picus & Odden, 2011). The choices policymakers and administrators make when allocating these funds are critical.

In recent years, calls for reforming PD have resulted in meaningful changes and important innovations, narrowing the size and scope of these activities to focus, for example, on the content or challenges of grade- or subject-specific teams. Some districts and schools are replacing independent providers of PD content with experienced teachers and instructional leaders with local expertise. Student work and assessment data are being injected into the discussion in new and innovative ways. However, the critical features of PD programs in most

public schools remain largely unchanged: they are generalized across teachers or teams, often abstracted from an individual teacher's own classroom context, and usually brief.

An emerging body of research suggests that coaching models of PD might provide a promising alternative organized around active-learning, job-embedded practice, and sustained focus. Recent evaluations of literacy coaching, math coaching, and web-based coaching focused on teacher-student relationships find mixed evidence on the efficacy of these programs. This study begins to build evidence on coaching models designed to improve behavior management and common instructional practices. Outside observers, principals, and students all rated teachers who received coaching as more effective than those who participated in standard PD activities provided by their schools although differences in principal ratings are not statistically significant at conventional levels. We also find suggestive evidence that these effects persist in the following academic year after teachers are no longer receiving coaching. Finally, the effect of MTC on our index of *Effective Teacher Practices* appears to be largely consistent across subjects, grade levels, and schools.

These results among the first cohort of participating teachers are most appropriately generalized to alternatively certified teachers who work with predominantly low-income minority students in urban charter schools and who are willing participants in a coaching program. This population of teachers is of substantial interest to policymakers given that one out of every four new teachers enters the profession through an alternative certification pathway.[5] The schools in our study are also representative of over two-thirds of the Recovery School District schools in Louisiana, as well as a growing number of schools in cities such as Washington, D.C., Philadelphia, and New York City. The relatively small variation in MTC

[5] Source: U.S. Department of Education, National Center for Education Statistics, Schools and Staffing Survey (SASS), "Public School Teacher Data File," 2011–12.

effect sizes across teachers and schools in the study provides additional support for the external validity of these findings amongst a similar population of teachers and schools. However, it remains an open question whether these effects would be realized if the program was taken to scale. This would require developing a larger coaching corps that is able to work effectively with teachers who might be less active participants in the coaching process. These results may also not be generalizable to coaching models that emphasize teacher reflection and self-assessment more than direct feedback.

Implications for Practice and Future Research

Districts interested in experimenting with teacher coaching need to find creative ways to address the challenges posed by the specialized personnel requirements and high costs of teacher coaching. We still know very little about what makes for an effective coach or what a system for selecting and training an effective corps of teacher coaches should look like. Differences in coach effectiveness is one possible explanation for the mixed evidence on teacher coaching. Coaching programs are also "among the most expensive approach to professional development" (Wayne et al., 2008, p. 470) because of their individualized and intensive nature. We estimate that MTC cost $9,000 per teacher, driven largely by personnel costs and a low teacher-to-coach ratio of 10-to-1. Further costs-benefit analyses will help to inform whether the relative returns to this costly investment are favorable compared to other less-costly approaches to professional development.

We propose a few ways in which districts, in partnership with researchers, could address and study these constraints. Broadly, districts may seek to develop a corps of coaches from within their current workforces. Such a strategy could have the added benefit of creating new career-ladder opportunities for expert teachers to serve as coaches. Given the high costs of

coaching, districts might be best served by experimenting with pilot programs that focus on novice and struggling teachers most in need of support.

Rapid advancements in 360-degree video capture and communication technology may also allow districts to harness expert support outside of the district and build more cost-effective coaching programs. As this technology improves and becomes more affordable, districts could invest in web-based coaching platforms like My Teaching Partner where teachers submit videos and receive individualized feedback from instructional experts online. This approach could increase the teacher-to-coach ratio by eliminating commuting costs and would more efficiently pair coaches with teachers in their grade-level and content areas of expertise. Far more research is needed in this area.

Future studies of MTC-style coaching programs should address the limitations of this research by evaluating larger and more diverse samples. We will extend these analyses by studying additional cohorts of New Orleans teachers as part of this multi-cohort study in order to examine the robustness and generalizability of these initial results. In addition, new studies should extend this work by implementing research designs that are optimized to estimate effects on student achievement in both the coaching year and follow-up year. The benefits of recruiting a diverse sample of teacher across K-12 grades and subjects in this study came at the cost of having a sufficient sample to examine effects on student achievement in tested grades and subjects. Many open questions remain about the potential value of coaching on general behavior management and instructional delivery skills. Answering these questions will take time, but the evidence-to-date suggests doing so will be a valuable investment.

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Tables

Table 1. Teacher Characteristics for Study Participants and Non-Participants and Treatment and Control Groups

	Means			Means		
	Teachers Not Selected to Participate in the Study	Teachers Selected to Participate in the Study	*p*-value on Difference between Participants and Non-Participants	Treatment Teachers	Control Teachers	*p*-value on Difference between Treatment and Control
Female	0.78	0.75	0.71	0.70	0.79	0.27
African-American	0.34	0.17	0.06	0.20	0.14	0.86
White	0.56	0.76	0.05	0.77	0.76	0.56
Age	-	26.1	-	26.1	26.1	0.96
Experience	3.28	3.97	0.14	3.93	4.00	0.89
First- or Second-Year Teacher	0.38	0.27	0.31	0.27	0.28	0.87
Third- or Fourth-Year Teacher	0.41	0.42	0.87	0.53	0.31	0.12
Fifth- or Higher-Year Teacher	0.22	0.31	0.38	0.20	0.41	0.11
Alternatively Certified	0.66	0.76	0.28	0.80	0.72	0.62
Master's Degree	-	0.22	-	0.20	0.24	0.75
College Institution Ranked Very Competitive or Higher	-	0.76	-	0.73	0.79	0.56
Interest in Coaching	8.16	9.11	0.00	9.23	8.98	0.32
F-statistic from Joint Test			2.18			0.58
p-value			0.06			0.81
n (teachers)	32	59		30	29	

Note: For non-participants, cells missing where data are not available. Joint tests for differences between participants and non-participants do not include Interest in Coaching, as teachers were selected on this variable. Treatment and control group means are estimated from regression models that control for randomization blocks. Joint tests include teachers' experience coded as a continuous variable and not as the three experience range indicators.

Table 2. Parameter Estimates of the Effect of MATCH Teacher Coaching on Measures of Teacher Effectiveness

	Coaching Year			Follow-Up Year
	Primary Findings	Impute Group Means	Multiple Imputation	Primary Findings
	(1)	(2)	(3)	(4)
Effective Teacher Practices Index	0.589*	0.526*	0.653*	0.476
	(0.240)	(0.204)	(0.267)	(0.364)
n (teachers)	52	59	59	33
MATCH Classroom Observation Rubric				
Achievement of Lesson Aim	0.579+	0.611*	0.658*	0.955**
	(0.311)	(0.258)	(0.296)	(0.307)
Behavioral Climate	0.663*	0.676*	0.739*	0.552
	(0.314)	(0.256)	(0.295)	(0.447)
n (teachers)	52	59	59	31
Principal Evaluation				
Principal Evaluation Composite	0.293+	0.134	0.297	0.240
	(0.168)	(0.171)	(0.238)	(0.39)
n (teachers)	52	59	59	33
TRIPOD Student Survey				
Control	0.092	0.113	0.142	-0.074
	(0.166)	(0.144)	(0.143)	(0.179)
Challenge	0.305**	0.261**	0.300**	0.183
	(0.113)	(0.093)	(0.094)	(0.234)
% Agree "learn a lot"	0.081*	0.065*	0.068+	0.103
	(0.034)	(0.028)	(0.036)	(0.080)
n (teachers)	50	59	59	33
n (students)	1414 to 1451	1763	1763	1001 to 1019

Notes: + p<0.1, *p<0.05, **p<0.01. Each cell contains results from a separate regression. All estimates, except the percent who agree that they "learn a lot" in their class, are reported as effect sizes with corresponding standard errors clustered by school in parentheses. All regressions include fixed effects for randomization blocks. The index of Effective Teacher Practices includes the five main outcome variables: the two observation items, the principal evaluation, and the two student survey domains. Imputation analyses account for missing data from seven teachers who dropped from the study and two teachers whose student surveys were lost in the mail. For teacher-level outcomes, we impute data for one observation for each missing teacher. For student-level outcomes, we impute data for the mean number of student observations by school level (23 for early elementary, 36 for upper elementary, and 38 for secondary). Parameters estimated with multiple imputation use all teacher characteristics in Table 1 and an indicator for treatment status to impute missing values across ten replication data sets.

Table 3. Estimates of Heterogeneity in Treatment Effects across Teacher Characteristics and Schools in Coaching Year

	Grade Level			Subject			School	
	Elementary (K-5)	Middle (6-8)	High (9-12)	All Subjects	Humanities	STEM	Standard Deviation of Treatment Effects	*p*-Value from Likelihood Ratio Test
Effective Teacher Practices Index	0.61+	0.587	0.343+	0.534	0.332	0.972+	0.039	0.992
	(0.328)	(0.601)	(0.196)	(0.395)	(0.326)	(0.526)		
n (teachers)	30	13	9	22	19	11	52	
MATCH Classroom Observation Rubric								
Lesson Aim	0.585	0.769	0.148+	0.563	0.348	0.948+	0.408	0.439
	(0.499)	(0.578)	(0.076)	(0.648)	(0.407)	(0.524)		
Behavioral Climate	0.738	0.671	0.345	0.743	0.214	1.041*	0.377	0.470
	(0.494)	(0.526)	(0.409)	(0.668)	(0.367)	(0.476)		
n (teachers)	30	13	9	22	19	11	52	
Principal Evaluation								
Principal Evaluation Composite	0.282	0.13	0.162	0.238	0.45	0.044	0.000	0.990
	(0.268)	(0.37)	(0.258)	(0.219)	(0.275)	(0.569)		
n (teachers)	30	13	9	22	19	11	52	
TRIPOD Student Survey								
Control	-0.044	0.063	0.276	-0.07	0.052	0.452+	0.449	0.008
	(0.141)	(0.238)	(0.527)	(0.184)	(0.313)	(0.264)		
Challenge	0.24*	0.286*	0.326	0.222*	0.178	0.612**	0.421	0.000
	(0.101)	(0.135)	(0.406)	(0.096)	(0.197)	(0.22)		
% Agree "learn a lot"	0.071	0.029	0.103	0.089	0.073	0.076	0.116	0.105
	(0.044)	(0.08)	(0.071)	(0.061)	(0.068)	(0.104)		

n (teachers)	29	12	7 to 9	22	16 to 18	10	50
n (students)	729 to 743	404 to 417	281 to 291	560 to 571	578 to 600	276 to 280	1414 to 1451

Notes: + p<0.1, *p<0.05, **p<0.01, ***p<0.001. Each cell contains results from a separate regression. For heterogeneity by grade level and subject, we report effect sizes by group with corresponding standard errors clustered by school in parentheses. All regressions include fixed effects for randomization blocks. The index of Effective Teacher Practices includes the five main outcome variables: the two observation items, the principal evaluation, and the two student survey domains. For heterogeneity across schools, we interact randomization blocks with a treatment indicator and report the standard deviation of school by treatment random effects and associated p-values estimated from likelihood ratio tests. Random effect models that allow for school-by-treatment effect heterogeneity with Challenge and % Agree "learn a lot" as outcomes do not converge. Estimates for these models are derived using fixed effects for randomization blocks interacted with treatment status.

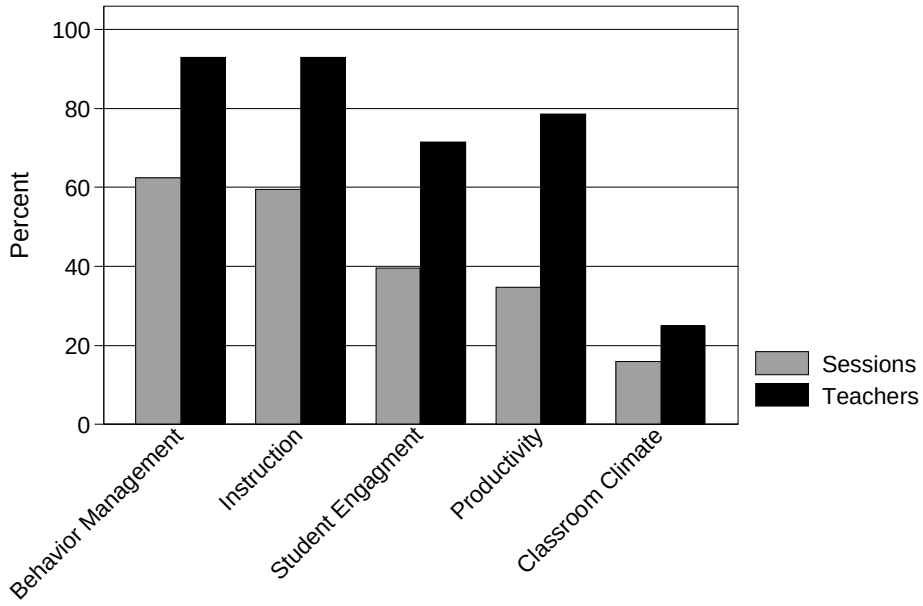
Table 4. Parameter Estimates of the Difference in Demographic Characteristics of Attritors Across Treatment and Control Groups.

	Coaching Year		Follow-Up Year	
	Coefficient	*p*-Value	Coefficient	*p*-Value
Female	0.019	0.961	-0.138	0.573
African-American	-0.048	0.888	-0.078	0.713
White	-0.336	0.388	0.001	0.996
Age	-3.274	0.361	-0.801	0.721
Experience	-1.346	0.492	-0.556	0.652
First- or Second-Year Teacher	0.100	0.805	0.352	0.160
Third- or Fourth-Year Teacher	0.098	0.825	-0.389	0.152
Fifth- or Higher-Year Teacher	-0.198	0.630	0.037	0.886
Alternatively Certified	0.123	0.751	-0.272	0.254
Master's Degree	-0.406	0.282	-0.238	0.304
College Institution Ranked Very Competitive or Higher	0.277	0.474	-0.169	0.485
Interest in Coaching	-0.807	0.358	-1.031	0.056

Notes: n= 59. In the coaching year, seven teachers were censored from the study, two from the treatment group and five from the control group. In the follow-up year, 26 teachers were censored from the study, nine from the treatment group and 17 from the control group.

Figures

Panel A



Panel B

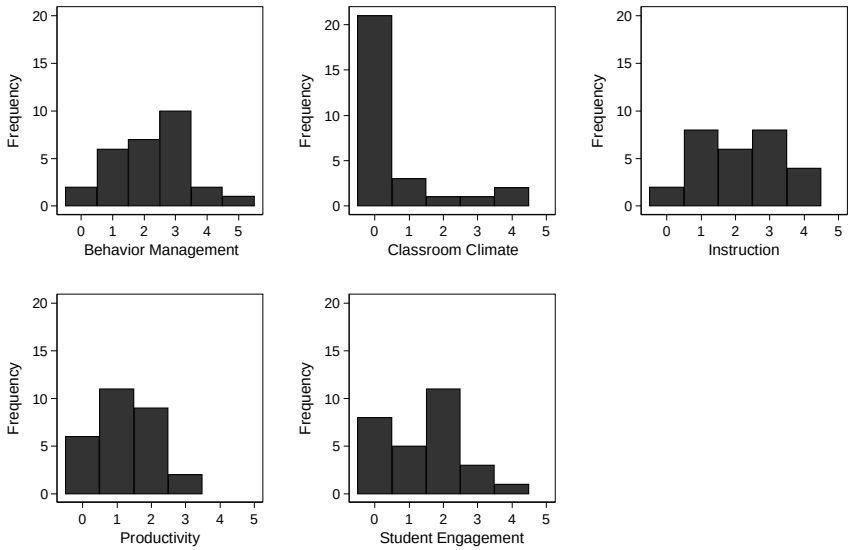


Figure 1. Panel A: Percent of sessions (n=101) where focus area was addressed (gray) and percent of teachers (n=28) who ever worked on given focus area (black). Panel B: Distributions of the number of sessions (n=101) that each teacher (n=28) worked on a given focus area.

Appendix A

MATCH Teacher Residency Program: External Evaluation Form

Teacher: Observer: Date: Class Period:

OVERALL ACHIEVEMENT OF LESSON AIM (1 TO 10 RATING): _____
(Degree to which the majority of students appeared to master the academic objective of the class.)

OVERALL BEHAVIORAL CLIMATE (1 TO 10 RATING): _____
(Degree to which you observed high student effort and low student misbehavior.

Assessing Aim Achievement

- More inference work involved in this assessment, as it's difficult to consistently collect and evaluate student work as a component of these observations.
- Informed by data from the Instructional Quality and Student Engagement categories of the rubric.

	Indicators of High ALA Ratings	Notes
Clarity and Rigor of the Aim	The aim of the lesson is clearly stated and often referred to. It's ambitious but achievable for the time allotted.	•
Alignment of Student Practice	The majority of the instruction – questions, tasks, explanations, etc – is aligned with the aim. Students are given sufficient "at-bats" to achieve the aim.	•
Assessment and Feedback	Student work – answers to questions, completed tasks, etc – is assessed throughout the lesson. Assessments provide students with multiple opportunities to get feedback. The standards for quality work are transparent.	•

Assessing Behavioral Climate

- The focus is on both student behavior and effort, which can generally be boiled down to time on task: *are students working hard to complete the tasks of a lesson with minimal distractions?*
- Informed by data from the Classroom Management and Student Engagement categories on the rubric.

	Indicators of High BC Ratings	Notes
Time on Task	Approximately 90% (or higher) of the students are consistently on task throughout the lesson. The students maintain their focus and effort irrespective of the task (e.g. listening to the teacher vs. taking notes vs. working in small groups, etc)	•
Transitions	Students frequently comply quickly with directions, resulting in transitions between tasks that occur with minimal loss of instructional time.	•
Response to Teacher Corrections	Students bounce back quickly after having their behavior corrected by the teacher. There's very minimal to no "pushback" after a correction, and the rest of the class maintains their effort/focus in these situations.	•

Explanation of Numerical Ratings:

Below Expectations = 1-3

Approaching Expectations = 4-5

Meets Expectations = 6-7

Exceeds Expectations = 8-10

*Priority Areas:

Objectives marked with an "X" are noted as priority areas for the teacher to improve upon his/her overall performance in subsequent evaluation

Appendix Tables

Appendix Table A1. Focus Areas of Coaching

Focus Area	Technique
Behavior Management	"Consequence Ladder" (Reward System)
	"Do it Agains" (Students Asked to Revise Behavior)
	"Narrating Compliance" (Reinforcing Positive Behavior)
	Developing an Authoritative Presence
	Movement around Room/Monitoring
	Reminder of Consequences
Classroom Climate	Private Corrections of Student Mistakes
	Smiling (teacher)
	Using a Positive Tone
Instructional Practices	Aligning Class Activities with Lesson Aim
	Backwards Planning
	Creating Measurable Objectives
	Developing Exit Tickets
	Increasing Rigor of Lesson through Higher-Order Thinking Tasks
	Increasing Time for and Scaffolding of Student Practice
	Utilizing Clear Directions
Productivity	Timing and Pacing of Lesson
Student Engagement	Call and Response
	Choral Response
	Cold Calling
	Decreasing Ratio of Teacher to Student Talk
	Rearranging Desks to Facilitate Group Conversation
	Turn and Talks (between students)

Appendix Table A2. Teacher Characteristics and Balance Between Participants and Non-Participants and Between Treatment and Control Groups in the Follow-up Year

	Means		*p*-value on Difference between Participants and Non-Participants in Follow-up Year	Means		*p*-value on Difference between Treatment and Control
	Teachers who Participated in the Follow-Up Year	Teachers Who Did Not Participate in the Follow-Up Year		Treatment Teachers in the Follow-up Year	Control Teachers in the Follow-up Year	
Female	0.79	0.69	0.41	0.78	0.81	0.72
African-American	0.21	0.12	0.33	0.24	0.16	0.91
White	0.76	0.77	0.92	0.76	0.76	0.91
Age	26.27	25.89	0.71	26.61	25.69	0.25
Experience	4.12	3.77	0.54	4.40	3.64	0.08
First- or Second-Year Teacher	0.24	0.31	0.58	0.18	0.36	0.72
Third- or Fourth-Year Teacher	0.49	0.35	0.29	0.56	0.35	0.29
Fifth- or Higher-Year Teacher	0.27	0.35	0.55	0.26	0.29	0.08
Alternatively Certified	0.73	0.81	0.48	0.77	0.65	0.37
Master's Degree	0.15	0.31	0.16	0.16	0.14	0.34
College Institution Ranked Very Competitive or Higher	0.76	0.77	0.92	0.80	0.69	0.40
Interest in Coaching	9.12	9.10	0.92	9.20	8.98	0.05
F-statistic from Joint Test			0.78			0.97
p-value			0.64			0.50
n (teachers)	33	26		21	12	

Note: Treatment and control group means are estimated from regression models that control for randomization blocks. Joint tests include teachers' experience coded as a continuous variable and not the three experience range indicators.